

Intermediate zero theorem and bisection method



Outline

Zeros of a function

Bisection method

Error estimates for the bisection method

Zeros of a function

Introduction: Zeros of a function

- ▶ In this section we study the problem of finding the **zeros** of a function, that is, the solutions of the equation $f(x) = 0$
- ▶ For most functions there are no solving formulas: we therefore need **existence** results and **approximation** methods
- ▶ We state and prove the **intermediate zero theorem** (Bolzano's theorem), one of the fundamental properties of continuous functions on an interval
- ▶ The proof is **constructive** and will give us an algorithm: the **bisection method**

Zeros of a function

Given a function f , we are interested in solving the equation:

$$f(x) = 0 \tag{1}$$

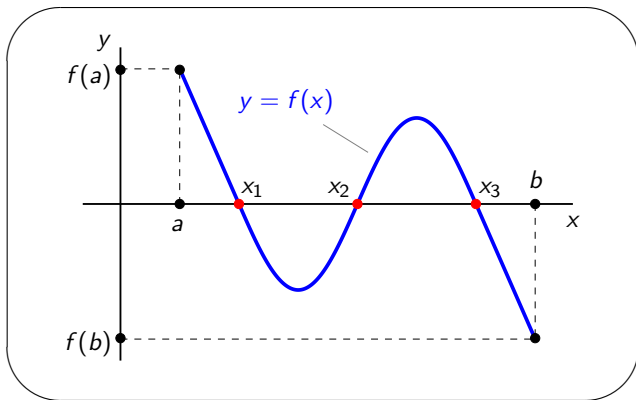
that is, in finding the **zeros** of f . The zeros are the solutions of equation (1), that is, the points c of the domain of the function where $f(c) = 0$.

Zeros of a function

- ▶ When f is a polynomial of degree ≤ 4 there are formulas that give the solutions of (1). However, if f is a polynomial of degree > 4 or a more complicated function, except in particularly lucky cases, there are no formulas for the solutions of equation (1).
- ▶ Geometrically, solving equation (1) means determining the x -coordinates of the intersection points between the graph of $y = f(x)$ and the x -axis. There can be: **infinitely many solutions, a finite number of solutions, no solutions.**

Example

Consider the function f with the following graph:



the equation $f(x) = 0$ has 3 solutions in the interval $[a, b]$, that is, the function f has 3 zeros, x_1 , x_2 and x_3 .

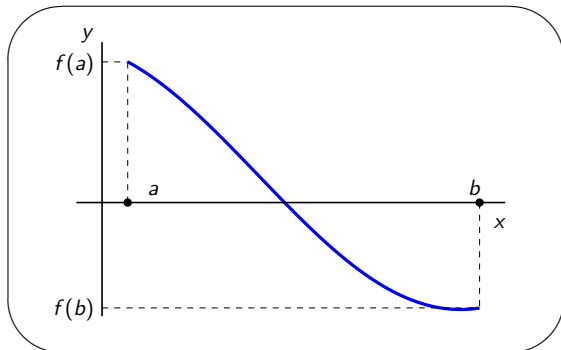
Intermediate zero theorem

Theorem 1: Intermediate zero theorem (Bolzano's theorem)

If a function $f : [a, b] \rightarrow \mathbb{R}$ is continuous on the interval $[a, b]$ and $f(a) \cdot f(b) < 0$, then there exists $c \in (a, b)$ such that $f(c) = 0$. If f is also strictly monotonic then the zero c is unique.

Idea of the proof (1/5)

- ▶ The idea of the proof is to construct two sequences that converge to the zero of the function. To understand how to construct them, consider for example a function with $f(a) > 0$ and $f(b) < 0$:

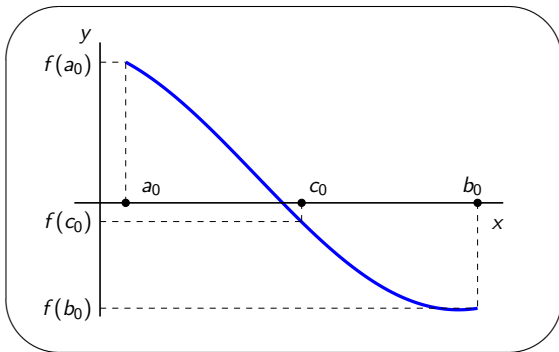


Idea of the proof (2/5)

- We set $a_0 = a$ and $b_0 = b$ and compute

$$c_0 = \frac{a_0 + b_0}{2}, \quad \text{the midpoint of the interval } [a_0, b_0].$$

In the previous example we have:



Idea of the proof (3/5)

If $f(c_0) = 0$ we have found a zero. If $f(c_0) \neq 0$ we look at the sign of $f(a_0) \cdot f(c_0)$ and proceed as follows:

$$\begin{cases} \text{if } f(a_0) \cdot f(c_0) < 0 \\ \text{if } f(a_0) \cdot f(c_0) > 0 \end{cases} \quad \text{we create } [a_1, b_1] \quad \begin{cases} \text{with } a_1 = a_0, \quad b_1 = c_0 \\ \text{with } a_1 = c_0, \quad b_1 = b_0 \end{cases}$$

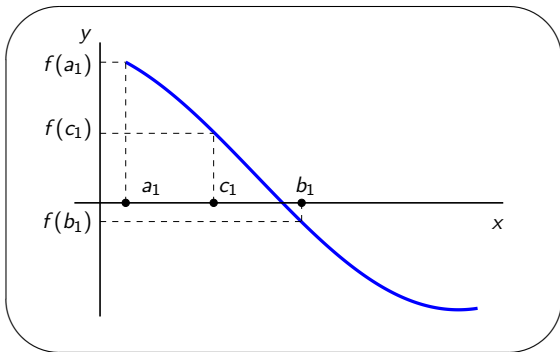
In the previous example we have $f(a_0) \cdot f(c_0) < 0$, hence we have $a_1 = a_0, \quad b_1 = c_0$.

Idea of the proof (4/5)

- With a_1 and b_1 , we compute:

$$c_1 = \frac{a_1 + b_1}{2}, \quad \text{the midpoint of the interval } [a_1, b_1].$$

In the previous example we have:



Idea of the proof (5/5)

If $f(c_1) = 0$ we have found a zero. If $f(c_1) \neq 0$ we look at the sign of $f(a_1) \cdot f(c_1)$ and proceed as follows:

$$\begin{cases} \text{if } f(a_1) \cdot f(c_1) < 0 \\ \text{if } f(a_1) \cdot f(c_1) > 0 \end{cases} \quad \text{we create } [a_2, b_2] \quad \begin{cases} \text{with } a_2 = a_1, \quad b_2 = c_1 \\ \text{with } a_2 = c_1, \quad b_2 = b_1 \end{cases}$$

In the previous example we have $f(a_1) \cdot f(c_1) > 0$, hence we have $a_2 = c_1, \quad b_2 = b_1$.

- We now generalize this idea in the proof of the theorem.

Proof (1/4)

Consider the following two recursively defined sequences:

$$a_0 = a, \quad a_{n+1} = \begin{cases} a_n & \text{if } f(a_n) \cdot f(c_n) < 0 \\ c_n & \text{if } f(a_n) \cdot f(c_n) > 0 \end{cases} \quad \forall n \in \mathbb{N}$$

$$b_0 = b, \quad b_{n+1} = \begin{cases} c_n & \text{if } f(a_n) \cdot f(c_n) < 0 \\ b_n & \text{if } f(a_n) \cdot f(c_n) > 0 \end{cases} \quad \forall n \in \mathbb{N}$$

where

$$c_n = \frac{a_n + b_n}{2}, \quad \forall n \in \mathbb{N}$$

Proof (2/4)

The two sequences create a sequence of intervals $[a_n, b_n]$ with the following properties:

- 1) We have $a_n \leq a_{n+1}$, hence the sequence $\{a_n\}$ is increasing and, since $a_n \leq b, \forall n$, it is also bounded. Moreover, we have $b_n \geq b_{n+1}$, hence the sequence $\{b_n\}$ is decreasing and, since $b_n \geq a, \forall n$, it is also bounded.
- 2) $b_n - a_n = \frac{b-a}{2^n}$ (each interval is half as long as the previous one)
- 3) $f(a_n) \cdot f(b_n) < 0$ (because of how a_n and b_n were chosen at each step)

Proof (3/4)

- By point 1), we can then deduce that the sequences $\{a_n\}$ and $\{b_n\}$ have a finite limit, thanks to the monotone sequence theorem. Hence:

$$a_n \rightarrow \ell_1 \in \mathbb{R} \quad \text{and} \quad b_n \rightarrow \ell_2 \in \mathbb{R} \quad \text{as} \quad n \rightarrow +\infty.$$

- From point 2) we deduce that:

$$b_n - a_n = \frac{b - a}{2^n} \rightarrow 0 \quad \text{as} \quad n \rightarrow +\infty,$$

$$\text{and therefore} \quad \ell_2 = \ell_1 = \ell$$

Proof (4/4)

By the continuity of f , we then have that:

$$f(a_n) \cdot f(b_n) \rightarrow (f(\ell))^2 \quad \text{as } n \rightarrow +\infty$$

On the other hand, from point 3) and the sign-preservation theorem we deduce $(f(\ell))^2 \leq 0$.

Therefore it must be $f(\ell) = 0$ and thus ℓ is the zero we are looking for, that is, $c = \ell$.



Sufficient, not necessary condition (1/2)

- We have proved:

$$\begin{aligned} f : [a, b] \rightarrow \mathbb{R} \text{ continuous on } [a, b] \text{ and } f(a) \cdot f(b) < 0 \\ \Rightarrow \text{there exists } c \in [a, b] \text{ such that } f(c) = 0 \end{aligned}$$

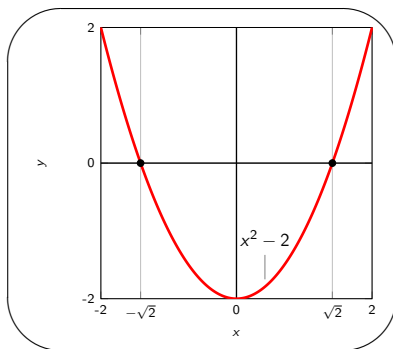
hence “ $f : [a, b] \rightarrow \mathbb{R}$ continuous on $[a, b]$ and $f(a) \cdot f(b) < 0$ ” is a sufficient condition for “there exists $c \in [a, b]$ such that $f(c) = 0$ ” and “there exists $c \in [a, b]$ such that $f(c) = 0$ ” is a necessary condition for “ $f : [a, b] \rightarrow \mathbb{R}$ continuous on $[a, b]$ and $f(a) \cdot f(b) < 0$ ”.

- The implication does not work in the other direction:

$$\begin{aligned} \text{there exists } c \in [a, b] \text{ such that } f(c) = 0 \\ \nRightarrow f : [a, b] \rightarrow \mathbb{R} \text{ continuous on } [a, b] \\ \text{and } f(a) \cdot f(b) < 0 \end{aligned}$$

Sufficient, not necessary condition (2/2)

It suffices, for example, to consider $f(x) = x^2 - 2$ on the interval $[-2, 2]$. We have $f(-2) = 2$ and $f(2) = 2$, hence $f(a) \cdot f(b) \neq 0$, but the function does have zeros:



Hence the theorem provides **sufficient** (but not necessary) conditions for the existence of a zero of a function.

The bisection method

- ▶ The proof of the theorem is constructive, that is, we have constructed two sequences that tend to a zero of the function (**constructive proof/algorithm**). This algorithm is called the **bisection method**.
- ▶ If the function f has several zeros in $[a, b]$, the procedure does not indicate which of them will be determined. The zero found depends on the interval $[a, b]$ given as input, and finding it may require infinitely many iterations. However, by stopping the procedure, as we will see, we obtain an estimate of the zero and of the error made.

Summary

Zeros of a function:

- ▶ The zeros of f are the solutions of $f(x) = 0$; in general there are no solving formulas

Intermediate zero theorem:

- ▶ f continuous on $[a, b]$ and $f(a) \cdot f(b) < 0 \Rightarrow$ there exists $c \in (a, b)$ with $f(c) = 0$ (unique if f is strictly monotonic)
- ▶ The proof constructs intervals $[a_n, b_n]$, halved at each step, with $f(a_n) \cdot f(b_n) < 0$, which converge to the zero
- ▶ The conditions are **sufficient** but not necessary (e.g. $x^2 - 2$ on $[-2, 2]$)

Bisection method

Introduction: Bisection method

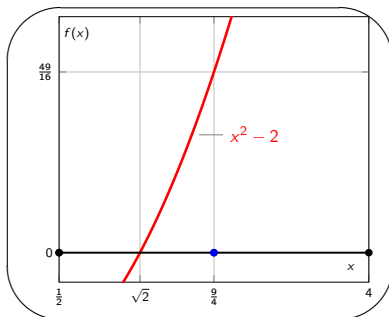
- ▶ In this section we apply the **bisection method** to a concrete example: the approximate computation of $\sqrt{2}$ as a zero of $x^2 - 2$
- ▶ We then study the **error estimate** obtained by stopping the method after n iterations
- ▶ We see how many iterations are needed to guarantee a given **tolerance** and we determine the **rate of convergence** of the method

Example (1/5)

- We look for a zero of the continuous function $f(x) = x^2 - 2$ on the interval $[\frac{1}{2}, 4]$, that is, we look for the value of $\sqrt{2}$. Initialization, with $n = 0$ we have:

$$[a_0, b_0] = [\frac{1}{2}, 4], \quad f(a) = -\frac{7}{4}, \quad f(b) = 14$$

$$\Rightarrow c_0 = \frac{9}{4} \quad \text{and} \quad f(c_0) = \frac{49}{16}$$

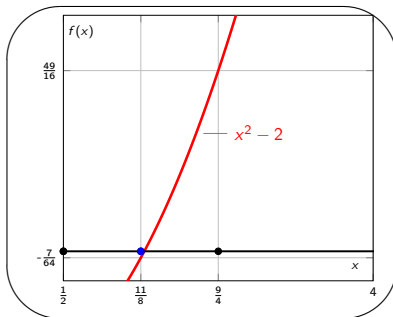


Example (2/5)

► Iteration $n = 1$, we have:

$$[a_1, b_1] = \left[\frac{1}{2}, \frac{9}{4}\right], \quad f(a_1) = -\frac{7}{4}, \quad f(b_1) = \frac{49}{16}$$

$$\Rightarrow c_1 = \frac{11}{8} \quad \text{and} \quad f(c_1) = -\frac{7}{64}$$

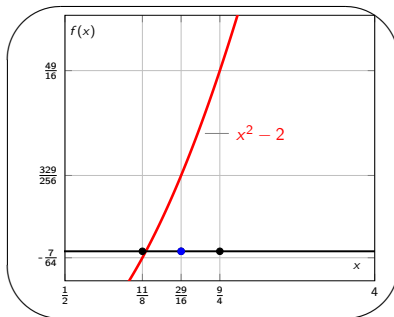


Example (3/5)

► Iteration $n = 2$, we have:

$$[a_2, b_2] = \left[\frac{11}{8}, \frac{9}{4}\right], \quad f(a_2) = -\frac{7}{64}, \quad f(b_2) = \frac{49}{16}$$

$$\Rightarrow c_2 = \frac{29}{16} \quad \text{and} \quad f(c_2) = \frac{329}{256}$$

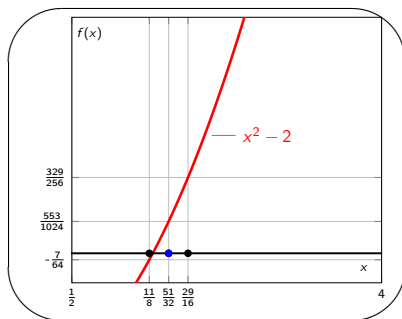


Example (4/5)

► Iteration $n = 3$, we have:

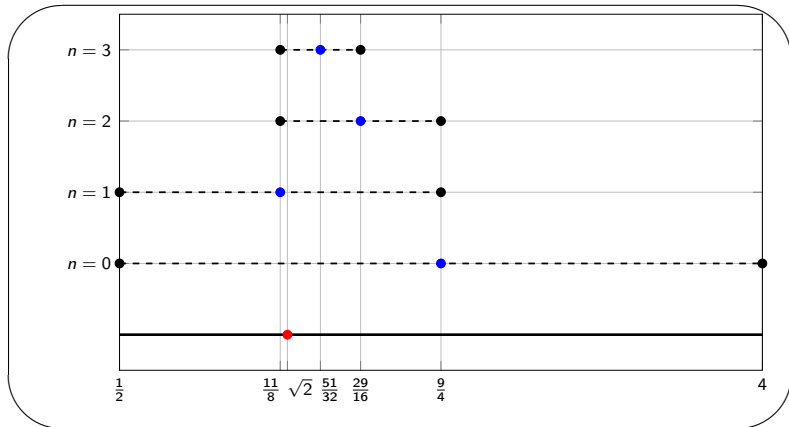
$$[a_3, b_3] = \left[\frac{11}{8}, \frac{29}{16} \right], \quad f(a_3) = -\frac{7}{64}, \quad f(b_3) = \frac{329}{256}$$

$$\Rightarrow c_3 = \frac{51}{32} \quad \text{and} \quad f(c_3) = \frac{553}{1024}$$



Example (5/5): intervals of the bisection method

Plotting the values obtained in the previous exercise, we have the following intervals and estimates of $\sqrt{2}$:



Error estimates for the bisection method

By stopping the method after n iterations we only obtain an estimate of the zero: we now see how to measure and control the error made.

Error estimates (1/3)

- At iteration n , we have:

$$c \in [a_n, b_n]$$

that is, the zero lies inside the n -th interval. Hence, stopping the procedure after n steps, a_n approximates the zero from below and b_n approximates the zero from above. Therefore:

$$a_n \leq c \leq b_n$$

- The estimate of the value c at iteration n is:

$$c_n = \frac{a_n + b_n}{2}$$

that is, the midpoint of the interval $[a_n, b_n]$.

Error estimates (2/3)

- ▶ The absolute error ε_n made at iteration n is:

$$\varepsilon_n = |c_n - c|$$

The value of c is unknown, hence we need a way to analyze the values of the errors that is independent of c .

Error estimates (3/3)

- ▶ We can estimate the error at iteration n as follows:

$$\varepsilon_n = |c_n - c| \leq \frac{b - a}{2^n}$$

this follows from:

- ▶ the length of the interval at iteration n is $\frac{b-a}{2^n}$
- ▶ the midpoint of this interval is c_n and c lies inside the interval itself
- ▶ the distance of any point c from the center of the interval is less than or equal to the length of the interval itself

If $n \rightarrow +\infty$ then $\varepsilon_n \rightarrow 0$, that is, the error tends to zero as n tends to infinity.

Improved error estimate

- At iteration n , we can improve the error estimate as follows:

$$\varepsilon_n = |c_n - c| \leq \frac{1}{2} \frac{b-a}{2^n} = \frac{b-a}{2^{n+1}}$$

that is, half the length of the interval $[a_n, b_n]$, since $c \in (a_n, c_n)$ or $c \in (c_n, b_n)$.

Example (1/2)

Error estimates for the bisection method

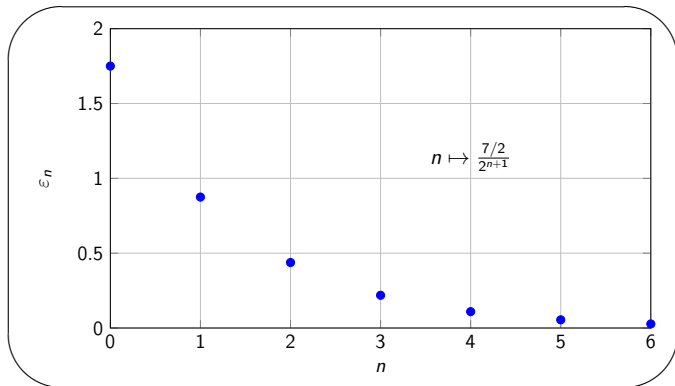
$$(\sqrt{2} = 1.414213562 \dots)$$

Let us go back to the previous exercise with $b - a = \frac{7}{2}$; in the first 6 iterations we have:

iteration	a_n	c_n	b_n	estimate of $\sqrt{2}$	ε_n
$n = 0$	$\frac{1}{2}$	$\frac{9}{4}$	4	2.25	$\leq \frac{7/2}{2^1} = 1.75$
$n = 1$	$\frac{1}{2}$	$\frac{11}{8}$	$\frac{9}{4}$	1.375	$\leq \frac{7/2}{2^2} = 0.875$
$n = 2$	$\frac{11}{8}$	$\frac{29}{16}$	$\frac{9}{4}$	1.8125	$\leq \frac{7/2}{2^3} = 0.4375$
$n = 3$	$\frac{11}{8}$	$\frac{51}{32}$	$\frac{29}{16}$	1.59375	$\leq \frac{7/2}{2^4} = 0.21875$
$n = 4$	$\frac{11}{8}$	$\frac{95}{64}$	$\frac{51}{32}$	1.484375	$\leq \frac{7/2}{2^5} = 0.109375$
$n = 5$	$\frac{11}{8}$	$\frac{183}{128}$	$\frac{95}{64}$	1.4296875	$\leq \frac{7/2}{2^6} = 0.0546875$

Example (2/2)

Behaviour of the maximum error estimate $\varepsilon_n \leq \frac{7/2}{2^{n+1}}$:



Number of iterations (1/3)

- In order to guarantee that the error made does not exceed a given tolerance δ , that is, to impose $\varepsilon_n \leq \delta$, we need to perform $n(\delta)$ iterations, where $n(\delta)$ is the smallest integer that satisfies the inequality:

$$n(\delta) > \log_2 \left(\frac{b-a}{\delta} \right) - 1$$

obtained by solving for $n(\delta)$ in the maximum error estimate:

$$\delta = \frac{b-a}{2^{(n(\delta)+1)}}$$

hence

$$2^{(n(\delta)+1)} = \frac{b-a}{\delta} \quad \text{and} \quad n(\delta) + 1 = \log_2 \left(\frac{b-a}{\delta} \right)$$

Number of iterations (2/3)

- ▶ Clearly, if $\delta \rightarrow 0$ then $n(\delta) \rightarrow +\infty$, and the smaller the error tolerance, the larger the number of iterations needed to guarantee it.
- ▶ If we wanted to reduce the tolerance by one decimal digit, that is, to go from δ to $\frac{\delta}{10}$, we would have:

$$n(\delta) \approx \log_2 \left(\frac{b-a}{\delta} \right) - 1 \quad \text{and}$$

$$n\left(\frac{\delta}{10}\right) \approx \log_2 \left(\frac{b-a}{\frac{\delta}{10}} \right) - 1$$

Number of iterations (3/3)

Since

$$\begin{aligned}\log_2 \left(\frac{b-a}{\frac{\delta}{10}} \right) - 1 &= \log_2 \left(\frac{b-a}{\delta} \cdot 10 \right) - 1 \\ &= \underbrace{\log_2 \left(\frac{b-a}{\delta} \right) - 1}_{\approx n(\delta)} + \log_2 10\end{aligned}$$

the number of additional iterations is independent of the interval $[a, b]$, and it is equal to

$$\log_2 10 \approx 3.32$$

Slow convergence

On average, more than three bisections are needed to improve the accuracy of the estimate by one significant digit;
convergence is therefore slow.

Example

Bisection method (error estimates)

Again for the previous example:

- ▶ if we want a tolerance (maximum error) $\delta = 0.00001 = 10^{-5}$:

$$n(10^{-5}) > \log_2 \left(\frac{7/2}{10^{-5}} \right) - 1 = 17.416 \dots$$

$$\text{hence } n(10^{-5}) = 18$$

- ▶ if we want a tolerance (maximum error) $\delta = 0.000001 = 10^{-6}$:

$$n(10^{-6}) > \log_2 \left(\frac{7/2}{10^{-6}} \right) - 1 = 20.738 \dots$$

$$\text{hence } n(10^{-6}) = 21$$

Rate of convergence (1/5)

We now want to understand how “fast” the error decreases by comparing the errors of two consecutive iterations. This estimate gives us information about the quality of the algorithm, that is, we want to determine the “**rate of convergence**” of the algorithm.

Rate of convergence (2/5)

- Consider the maximum error, which we denote by ε_n^* ; we have:

$$\varepsilon_n^* = \frac{b-a}{2^{n+1}} \quad \text{and} \quad \varepsilon_n \leq \varepsilon_n^*$$

hence

$$\frac{\varepsilon_{n+1}^*}{\varepsilon_n^*} = \frac{1}{2} \quad \text{since} \quad \frac{\frac{b-a}{2^{n+2}}}{\frac{b-a}{2^{n+1}}} = \frac{1}{2}$$

that is, the maximum error is halved at each iteration.
Consequently, we have:

$$\underbrace{\varepsilon_{n+1}^*}_{\geq \varepsilon_{n+1}} = \frac{1}{2} \underbrace{\varepsilon_n^*}_{\geq \varepsilon_n}$$

Rate of convergence (3/5)

The maximum absolute error of the bisection method at each step is proportional to the maximum absolute error at the previous step. **The rate of convergence is linear.**

Rate of convergence (4/5)

- For every triple c_{n+1} , c_n and c_{n-1} , with $n \geq 1$, we have the following relation:

$$c_{n+1} = \frac{c_{n-1} + c_n}{2} \quad (2)$$

that is, the estimate c_{n+1} is the midpoint of the interval with endpoints c_{n-1} and c_n . We now write:

$$c_{n+1} = c + \varepsilon_{n+1}, \quad c_n = c + \varepsilon_n, \quad c_{n-1} = c + \varepsilon_{n-1}$$

Rate of convergence (5/5)

- Now, substituting into equation (2), we obtain:

$$\begin{aligned}c + \varepsilon_{n+1} &= \frac{c + \varepsilon_n + c + \varepsilon_{n-1}}{2} = \frac{2c + \varepsilon_n + \varepsilon_{n-1}}{2} \\&= c + \frac{\varepsilon_n + \varepsilon_{n-1}}{2}\end{aligned}$$

hence

$$\varepsilon_{n+1} = \frac{\varepsilon_n + \varepsilon_{n-1}}{2} = \frac{1}{2}\varepsilon_n \left(\frac{\varepsilon_{n-1}}{\varepsilon_n} + 1 \right)$$

- The term $\varepsilon_{n-1}/\varepsilon_n$, that is, the ratio between the errors of two consecutive iterations, can be an arbitrarily small or large number (indeed, the error at each iteration can increase or decrease).

Summary

Bisection method:

- ▶ At each step the interval $[a_n, b_n]$ is halved while keeping $f(a_n) \cdot f(b_n) < 0$; the estimate of the zero is $c_n = \frac{a_n + b_n}{2}$

Error estimates:

- ▶ $\varepsilon_n = |c_n - c| \leq \frac{b-a}{2^{n+1}}$, independent of the (unknown) zero c
- ▶ Tolerance δ : $n(\delta)$ is the smallest integer with $n(\delta) > \log_2 \left(\frac{b-a}{\delta} \right) - 1$
- ▶ Each additional decimal digit costs about $\log_2 10 \approx 3.32$ iterations: **slow**, **linear** convergence

Final summary

- ▶ The **zeros** of a function are the solutions of $f(x) = 0$, often not computable with explicit formulas
- ▶ **Intermediate zero theorem**: a continuous function on $[a, b]$ with $f(a) \cdot f(b) < 0$ has at least one zero in (a, b) ; the conditions are sufficient but not necessary
- ▶ The proof is constructive and gives the **bisection method**
- ▶ The maximum error $\frac{b-a}{2^{n+1}}$ is halved at each iteration: convergence is **linear** and about 3.32 iterations are needed for each decimal digit